TEXT ANALYTICS PROJECT ON PUBLIC DATA

Finding recurring problems in free text complaints

84,000 Ford owner complaints from the US road safety regulator (NHTSA), analysed as if I were the manufacturer's field quality team.

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THE QUESTION

Category codes miss much of what free text complaints say. Which recurring problems matter most, which are growing, and which are parts or process problems rather than product defects?

83,992
Ford complaints, January 2020 to September 2026; all analysed, no sampling

68
named recurring problems, covering 90.3% of all complaints

69%
assigned to exactly the right problem in a manual check of 200; 93% right or closely related

9.3%
are owners waiting for recall parts or repairs, the fastest growing complaint type

METHOD: SIX STEPS, ALL RUNNING OFFLINE ON A LAPTOP

1 Load
NHTSA flat files, tab delimited, no header row. Personal fields (operator name, VIN, dealer details) are dropped on load. Complaints and component codes go into separate tables.

2 Clean
A third of complaints are written by hotline staff from a template. Left in, texts cluster by who wrote them, not by what broke, so the template is stripped. Model names are masked.

3 Embed and cluster
Sentence embeddings (all-MiniLM-L6-v2), UMAP and HDBSCAN find 89 clusters. Unclustered complaints join their nearest cluster if similarity is at least 0.60.

4 Label
A local LLM (Qwen 3.5 4B on Ollama) reads 12 examples per cluster and returns a label, vehicle system and issue type as structured JSON.

5 Review
Every LLM label checked by hand: 41 of 89 corrected, for example a brake defect the model had marked as a recall issue. Clusters merged into 68 problems.

6 Measure and report
Manual accuracy check on 200 random complaints. Power BI star schema with a Pareto of problems, severity weighting and the 12 month change per problem.

WHAT CAME OUT

Top 5 recurring problems	Complaints	Crash, fire or injury	vs prior 12 months
Coolant leaks into cylinders, engine needs replacing	8,920	0.9%	-41%
Power steering assist fails while driving	7,041	2.3%	-44%
Transmission slips, shifts hard or fails	4,796	0.9%	-1%
Brake pedal sinks to floor, brake fluid leak	3,677	4.8%	+21%
Transmission unexpectedly downshifts	3,633	0.9%	-10%

- **Parts, not engineering.** 9.3% of complaints are owners waiting for recall parts or repairs, and this is growing fastest: backup camera recall parts +83%, delayed recall repairs +75%. It needs a different owner than a design fault.
- **Brakes are the one to watch:** up 21%, with the highest crash, fire or injury share in the top 15.
- **The text routes what the codes cannot.** 5,306 complaints carry only the code "UNKNOWN OR OTHER"; 86% of them were assigned to a named problem.

HOW ACCURATE IS IT?

Manual check, 200 random complaints	Exactly right	Right or close	Wrong
Placed inside a cluster	72%	95%	5%
Joined the nearest cluster	62%	89%	11%
Weighted overall	**69%**	**93%**	**7%**

Most "close" cases are overlapping problems, such as two transmission problems that blur into each other. Raising the 0.60 cutoff to 0.65 would roughly halve the wrong share among nearest matches but drop coverage from 90% to about 84%; for a volume report, coverage won.

WHY IT MATTERS FOR QUALITY AND SERVICE TEAMS

- The same pipeline fits **warranty claims, service reports, field tickets and customer feedback**.
- It finds recurring issues that **category codes miss**, and shows what is **growing**, not only what is big.
- It separates **product defects from parts and process delays**.
- It runs **fully offline**, so confidential data stays on the company's machines.

LIMITS OF PUBLIC DATA, AND HOW I HANDLED THEM

No cost or hours per case
Effort is severity weighted volume (injury 10, crash or fire 5, towed 2, other 1); real cost data plugs straight in.

No customer ID
A repeat is the same problem in one model and year, month after month; real data would use the serial number.

Self reported complaints
Unverified owner reports: the results show where to look, not proven root causes.

Code, method and full results
github.com/KerenyiKrisz/nhtsa-complaint-clustering

Data
NHTSA Office of Defects Investigation, complaints flat files, nhtsa.gov

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